A First Course in Causal Inference Ch. 10

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Ch. 10 Observational Studies, Selection Bias, and Nonparametric Identification of Causal Effects

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Cochran and Chambers (1965) summarized two commonn characteristics of observational studies:

1. the objective is to elucidate cause-and-effect relationships;
2. it is not feasible to use controlled experimentation.

- This part will discuss causal inference with observational studies.
- Many ideas of causal inference with observational studies are deeply connected to those with randomized experiments.

10.2 Causal effects and selection bias under the potential outcomes framework

- For unit i ($i = 1, \ldots, n$), we have pretreatment covariates X_i, a binary treatment indicator Z_i, and an observed outcome Y with two potential outcomes $Y_i(1)$ and $Y_i(0)$.
- We assume $\{X_i, Z_i, Y_i(1), Y_i(0)\}_{i=1}^{n} \overset{\text{i.i.d.}}{\sim} \{X, Z, Y(1), Y(0)\}$.
- The causal effects of interest are the average causal effect $\tau = \mathbb{E}[Y(1) - Y(0)]$, the average causal effect on the treated units $\tau_T = \mathbb{E}[Y(1) - Y(0) \mid Z = 1]$, and the average causal effect on the control units $\tau_C = \mathbb{E}[Y(1) - Y(0) \mid Z = 0]$.

10.2 Causal effects and selection bias under the potential outcomes framework

Rewrite the average causal effects on the treated units and control units:

$$\tau_T = \mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 1] = \underbrace{\mathbb{E}[Y \mid Z = 1]}_{\text{observable}} - \underbrace{\mathbb{E}[Y(0) \mid Z = 1]}_{\text{unobservable}}.$$

$$\tau_C = \mathbb{E}[Y(1) \mid Z = 0] - \mathbb{E}[Y(0) \mid Z = 0] = \underbrace{\mathbb{E}[Y(1) \mid Z = 0]}_{\text{unobservable}} - \underbrace{\mathbb{E}[Y \mid Z = 0]}_{\text{observable}}.$$

These unobservable terms are *counterfactuals*.

10.2 Causal effects and selection bias under the potential outcomes framework

- The simple difference in means, known as the *prima fucie* causal effect

$$\tau_{\mathsf{PF}} = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0] = \mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0]$$

  is generally biased for the causal effects defined above.
- For example, $\tau_{\mathsf{PF}} - \tau_T = \mathbb{E}[Y(0) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0]$ is not zero in general.
- The equation above quantifies *selection bias*.
- The reason why randomization matters is because the *selection bias* term is zero.
- Under the CRE, $\tau_{\mathsf{PF}} = \tau_T = \tau_C = \tau$.
- The fundamental benefit of randomization is to balance the distributions of the potential outcomes across the groups.

10.3 Sufficient conditions for nonparametric identification of causal effects

- Causal inference with observational studies is challenging and it relis on strong assumptions.
- A strategy is to use the information from the pretreatment covariates and assume that conditioning on observed covariates X, the selection bias terms are zero.

$$\mathbb{E}[Y(0) \mid Z = 1, X] = \mathbb{E}[Y(0) \mid Z = 0, X], \quad \mathbb{E}[Y(1) \mid Z = 1, X] = \mathbb{E}[Y(1) \mid Z = 0, X]$$

- This assumption states that the differences in the means of the potential outcomes across the treatment and control groups are entirely due to the difference in the observed covariates.
- The average causal effect τ is nonparametrically identifiable under this assumption.

10.3 Sufficient conditions for nonparametric identification of causal effects

Definition 10.1 (identification)

A parameter θ is identifiable if it can be writte as a function of the distribution of the observed data under certain model assumptions. A parameter θ is non-parametrically identifiable if it can be written as a function of the distribution of the observed data without any parametric model assumptions.

- This assumption is fundamental in causal inference with observational studies.
- The parameter of interest $\tau = \mathbb{E}[Y(1) - Y(0)]$ depends on some unobserved random variables, so it is unclear whether it is nonparametrically identifiable based on observed data.
- Because τ_{PF} depends only on the observables, it is nonparametrically identified by definition.

10.3 Sufficient conditions for nonparametric identification of causal effects

Theorem 10.1

Under the assumption, the average causal effect τ is identified by

$$\begin{aligned}\tau &= \mathbb{E}[\tau(X)] \\ &= \mathbb{E}[\mathbb{E}[Y \mid Z = 1, X]] - \mathbb{E}[Y \mid Z = 0, X] \\ &= \int \{\mathbb{E}[Y \mid Z = 1, X = x] - \mathbb{E}[Y \mid Z = 0, X]\} f(x) dx\end{aligned}$$

10.3 Sufficient conditions for nonparametric identification of causal effects

Usually we impose the strong assumption:

Assumption 10.1 (ignorability)

We have $Y(z) \perp\!\!\!\perp Z \mid X \quad (z = 0, 1)$.

Assumption 10.2 (strong ignorability)

We have $\{Y(1), Y(0)\} \perp\!\!\!\perp Z \mid X$.

These assumptions are not necesarry for identifying τ but they cannot be relaced if the parameter of interest is the causal effects on other scales.

10.3 Sufficient conditions for nonparametric identification of causal effects

- Intuitively, the assumption rules out all unmeasured covariates that affect the treatment and outcome simultaneously.
- The ignorability assumption can be reasonable if we observe a rich set of covariates X that affect the treatment and the outcome simultaneously.
- It is fundamentally untestable.
- We may justify it based on the scientific background knowledge but we are often not sure whether it holds or not.

10.4 Two simple estimation strategies and their limitations

- If the covariate $X_i \in \{1, \ldots, K\}$ is discrete then the ignorability assumption reads as $Y(z) \perp\!\!\!\perp Z \mid X = k$.
- Therefore, we can use the estimator $\hat{\tau} = \sum_{k=1}^{K} \pi_{[k]} \{\hat{\bar{Y}}(1) - \hat{\bar{Y}}(0)\}$, where $\pi_{[k]}$ denotes a propensity score.
- This method is widely used but there are several problems:
- First, it works well for small K, but it leads to the poorly defines $\hat{\tau}_{[k]}$ for large K because some strata are very likely to have $n[k]1 = 0$ or $n_{[k]1} = 0$.
- Second, it is not obvious how to apply this stratification method to multidimensional continuous or mixed covariates X.

10.4 Two simple estimation strategies and their limitations

- The most commonly used method is to run the OLS with an additive model of the observed outcome on the treatment indicator and covariates: $\mathbb{E}[Y \mid Z, X] = \beta_0 + \beta_z Z + \beta_x^\top X$.
- If the linear model is correct, then we have

$$\tau(X) = \mathbb{E}[Y \mid Z = 1, X] - \mathbb{E}[Y \mid Z = 0, X] = \beta_z.$$

- This estimator implies that the causal effect is homogeneous with respect to the covariates.
- Thie, coupled with ignorability, implies that $\tau = \mathbb{E}[\tau(X)] = \beta_z$.
- If ignorability holdes and the outcome model is linear, then the average causal effect equals the coeff. of Z.
- At the same time, this is valid only under two strong assumptions: *ignorability* and *the linear model*.

10.4 Two simple estimation strategies and their limitations

- If we incorporate heterogenous causal effects, we include theinteraction term:

$$\mathbb{E}[Y \mid Z, X] = \beta_0 + \beta_z Z + \beta_x^\top X + \beta_{zx}^\top XZ.$$

- We then have

$$\tau(X) = \mathbb{E}[Y \mid Z = 1, X] - \mathbb{E}[Y \mid Z = 0, X] = \beta_z + \beta_{zx}^\top X.$$

- With ignorability, it implies that

$$\tau = \mathbb{E}[\tau(X)] = \mathbb{E}[\beta_z + \beta_{zx}^\top X] = \beta_z + \beta_{zx}^\top \mathbb{E}[X].$$

- The estimator for τ is then $\hat{\beta}_z + \hat{\beta}_{zx}^\top \bar{X}$.

10.4 Two simple estimation strategies and their limitations

- If we center the covariates to ensure $\bar{X} = 0$, then the estimator is simpli the regression coeff. of Z.
- In general, we can use other more complex models to estimate the causal effects.
- If we build two predictors $\hat{\mu}_1(X)$ and $\hat{\mu}_0(X)$ based on the treated and control data, respectively.
- Then we have an estimator for CATE: $\hat{\tau}(X) = \hat{\mu}_1(X) - \hat{\mu}_0(X)$, and an estimator for ATE: $\hat{\tau}^{\text{reg}} = n^{-1} \sum_{i=1}^{n} \{\hat{\mu}_1(X) - \hat{\mu}_0(X)\}$.
- It is sometimes called the *outcome regression* estimator.

10.4 Two simple estimation strategies and their limitations

- The biggest problem with the above approach based on outcome regressions is its sensitivity to the specification of the outcome model.
- Researchers migh report their favorable causal effects estimates after searching over a wide set of candidate models, without confessing the model searching process.
- This is a major source of *p*-hacking in causal inference.

References I

Cochran, W. G. and S. P. Chambers (1965). The planning of observational studies of human populations. *Journal of the Royal Statistical Society. Series A (General) 128*(2), 234–266.